

WPW Wide Complex Tachycardia

Section 1: Case Summary

Scenario Title:	WPW Wide Complex Tachycardia
Keywords:	Tachycardia, Wide complex, ACLS, WPW, cardioversion, AV nodal blocker
Brief Description of Case:	A 37F with wide complex tachycardia (wolfe-parkinson-white) presents, initially stable, and after unsuccessful treatments, decompensates to unstable atrial fibrillation, and requires synchronized cardioversion.

Goals and Objectives	
Educational Goal:	Recognition of wide complex tachycardia
Objectives: (Medical and CRM)	1. Recognize wide complex tachycardia 2. Follow the ACLS tachycardia (stable and unstable) algorithm 3. Communicate clearly with team during a resuscitation including closed loop communication
EPAs Assessed:	C1: Resuscitating and coordinating care for critically ill patients

Learners, Setting and Personnel			
Target Learners:	☒ Junior Learners	☒ Senior Learners	☐ Staff
	☐ Physicians	☐ Nurses	☐ RTs
	☐ Other Learners:		
Location:	☒ Sim Lab	☐ In Situ	☐ Other:
Recommended Number of Facilitators:	Instructors: 1 - 2		
	Sim Actors:		
	Sim Techs: 1		

Scenario Development	
Date of Development:	May 25, 2022
Scenario Developer(s):	Kelly Huang, PGY-3
Affiliations/Institutions(s):	University of British Columbia, Department of Emergency Medicine
Contact E-mail:	Kelly.huang@alumni.ubc.ca
Last Revision Date:	
Revised By:	
Version Number:	



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Section 2A: Initial Patient Information

A. Patient Chart					
Patient Name: Vanessa		Age: 32		Gender: F	Weight: 70kg
Presenting complaint: Palpitations					
Temp: 37.1	HR: 185	BP: 115/75	RR: 22	O ₂ Sat: 97%	FiO ₂ : RA
Cap glucose: 5.1			GCS: 15		
Triage note: 37F with 3 hour history of palpitations. It started when she was driving her kids to school and has gotten worse. She denies any recreational drug use. She denies any cardiac history.					
Allergies: NKDA					
Past Medical History: Depression			Current Medications: Escitalopram		

Section 2B: Extra Patient Information

A. Further History	
1. No recent long-haul flights or immobilization 2. No previous DVT or PE 3. No active cancer, hemoptysis, or leg swelling 4. Family history is non contributory	
B. Physical Exam	
Cardio: tachycardia, no murmurs	Neuro: GCS 15, no focal deficits
Resp: mild respiratory distress, clear to bases	Head & Neck: normal
Abdo: soft, non tender	MSK/skin: normal
Other: looks uncomfortable and complains that it feels weird	

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Section 3: Technical Requirements/Room Vision

A. Patient
☒ Mannequin (<i>specify type and whether infant/child/adult</i>)
☐ Standardized Patient
☐ Task Trainer
☐ Hybrid
B. Special Equipment Required
Crash cart with defibrillator
C. Required Medications
Procainamide, ketamine, fentanyl Possibly adenosine, amiodarone, sotalol
D. Moulage
None required
E. Monitors at Case Onset
☒ Patient on monitor with vitals displayed
☐ Patient not yet on monitor
F. Patient Reactions and Exam
None

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Section 4: Sim Actor and Standardized Patients

None needed



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Section 5: Scenario Progression

Scenario States, Modifiers and Triggers				
Patient State/Vitals	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State		Facilitator Notes
1. Baseline State Rhythm: regular wide complex tachycardia HR: 180 BP: 125/85 RR: 22 O ₂ SAT: 97 % T: 37.1°C GCS: 15	Alert, uncomfortable, responsive	<u>Expected Learner Actions</u> ☐ Order ECG ☐ Obtain IV access ☐ Apply cardioresp monitors and defib pads ☐ Focused exam ☐ Focused history ☐ Recognize wide complex tachycardia ☐ Attempt chemical cardioversion (procainamide)	<u>Modifiers</u> - show ECG #1 when ordered - show CXR & labs when ordered <u>Triggers</u> - AV nodal blocker → phase 2 - Procainamide/amiodarone → phase 3 - Electrical cardioversion → patient refuses - After 4 minutes → phase 2	If team elects to do an electrical cardioversion, patient will refuse "That sounds unpleasant, can we try something else?"
2. Polymorphic VT Rhythm: irregular polymorphic ventricular tachycardia HR: 200 BP: 88/60 RR: 22 O ₂ : 97% GCS: 14	Drowsy, eyes half closed	<u>Expected Learner Actions</u> ☐ Call for help ☐ Repeat set of vitals ☐ Perform synchronized cardioversion ☐ Consider light hemodynamically neutral sedation/analgesia	<u>Modifiers</u> - if 2 mins pass and rhythm change not noticed, RN to prompt "it seems the rhythm looks different now" <u>Triggers</u> - Synchronized cardioversion -> phase 3	
3. Hypotension Rhythm: regular wide complex tachycardia HR: 180 BP: 88/60 RR: 22 O ₂ : 97% GCS: 14	Eyes half open, not as responsive	<u>Expected Learner Actions</u> ☐ Call for help if not already done ☐ Discontinue antiarrhythmic infusion ☐ Recognize persistent hypotension (repeat vitals) ☐ Synchronized cardioversion	<u>Triggers</u> - Synchronized cardioversion -> phase 4	Hypotension persists from previous antiarrhythmic, prompting repeat electrically cardiovert
4. Stable	Responsive	<u>Expected Learner Actions</u>	<u>Modifiers</u>	- Show ECG #3



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Rhythm: sinus with delta waves HR: 82 BP: 122/90 RR: 16 O2: 97% GCS: 15		☐ Order repeat ECG and repeat vital signs ☐ Consult cardiology for further guidance of management and workup	- Show ECG #3 once ordered <u>Triggers</u> - Cardiology pages back and end case	
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Simulation Scenario Template

Appendix A: Laboratory Results

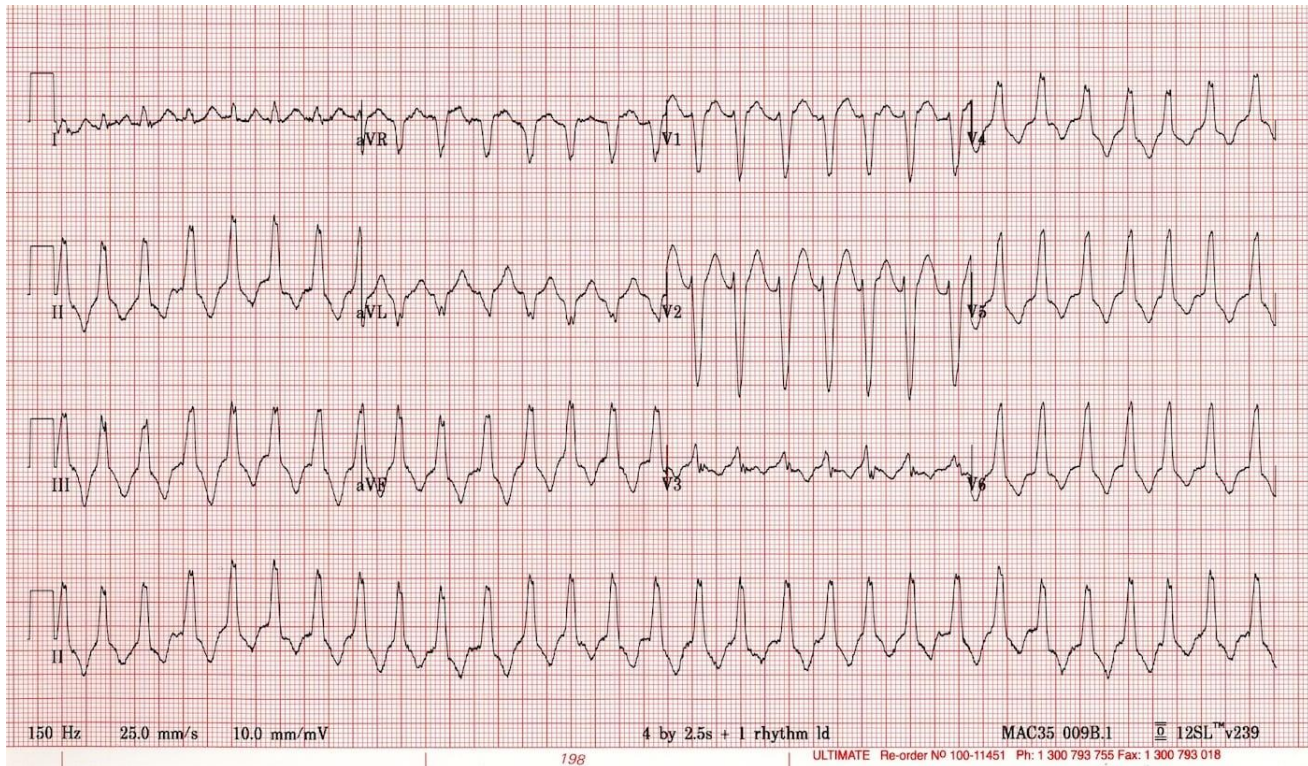
<u>CBC</u> WBC 6.0 Hgb 132 Plt 310 <u>Lytes</u> Na 141 K 4.0 Cl 108 HCO ₃ Cr 80 Glucose 6.1	<u>Cardiac/Coags</u> hsTrop 8 D-dimer 210 INR 1.1 aPTT 37 <u>Other</u> B-HCG neg
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Simulation Scenario Template

Appendix B: ECGs, X-rays, Ultrasounds and Pictures

ECG #1

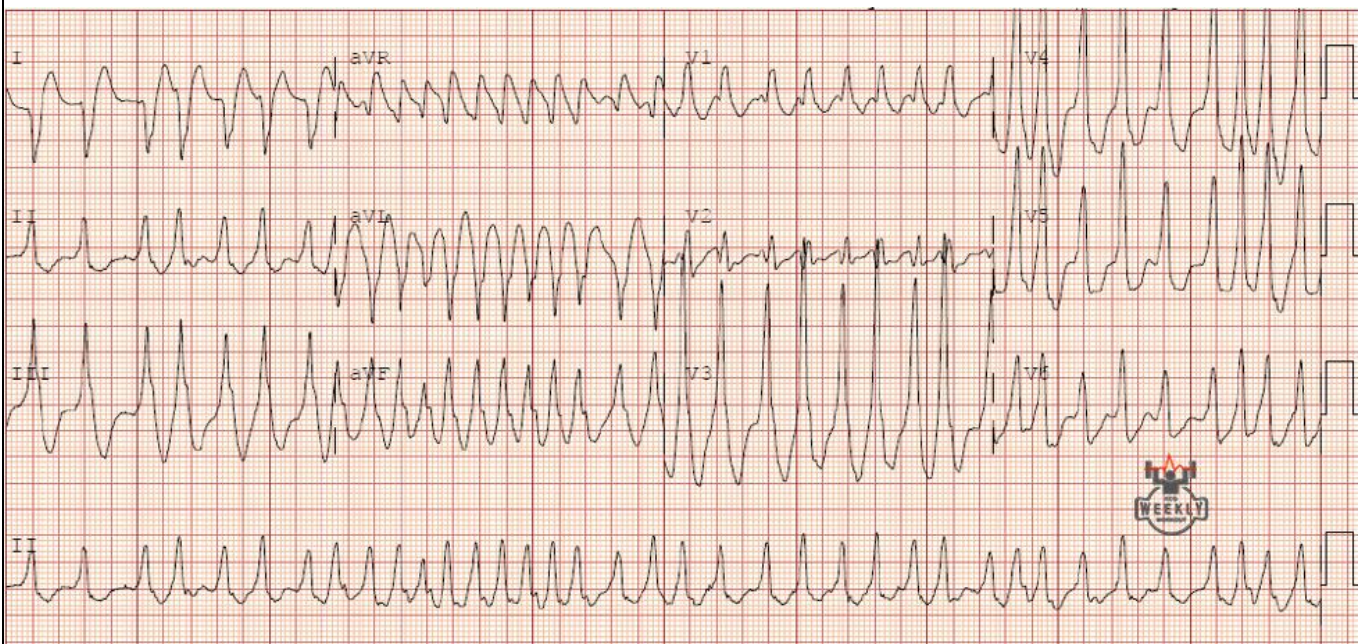
Wide complex regular tachycardia (antidromic AVRT)



Source: <https://litfl.com/wp-content/uploads/2019/04/WCTfeb2012-1.jpg>

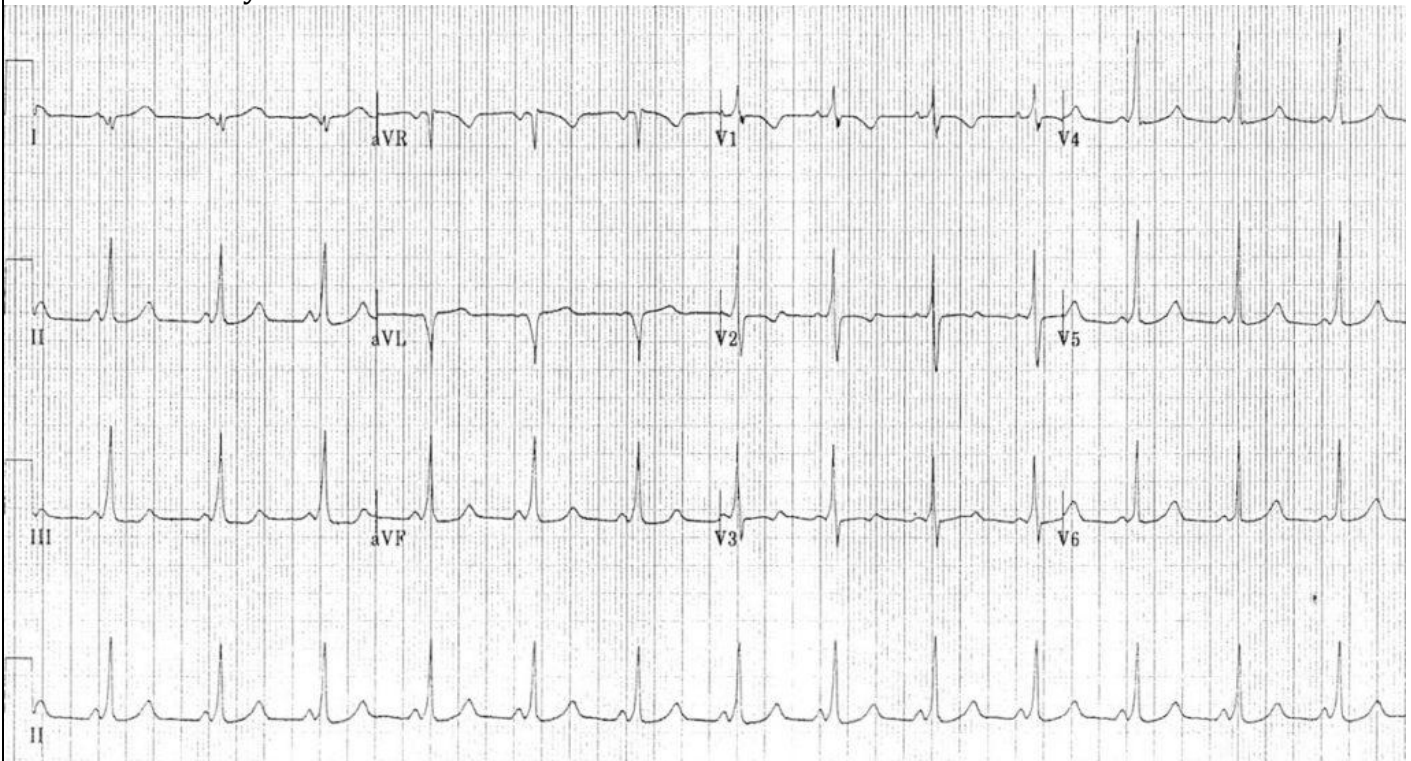
Simulation Scenario Template

ECG#2: Irregular, polymorphic wide complex tachycardia (afib with WPW)



Source: <https://www.acc.org/education-and-meetings/patient-case-quizzes/2020/03/09/09/00/treatment-of-wide-complex-tachycardias#>

ECG #3: sinus rhythm with WPW



Source: <https://litfl.com/wp-content/uploads/2018/08/Orthodromic-AVRT-post-adenosine-1-2.jpg>

Simulation Scenario Template

Tachycardia With a Pulse Algorithm



Assess appropriateness for clinical condition.
Heart rate typically $\geq 150/\text{min}$ if tachyarrhythmia.

Identify and Treat Underlying Cause

- Maintain patient airway; assist breathing as necessary
- Oxygen as indicated
- Cardiac monitor to identify rhythm; monitor blood pressure and oximetry

Persistent Tachyarrhythmia Causing:

- Hypotension?
- Acutely altered mental status?
- Signs of shock?
- Ischemic chest discomfort?
- Acute heart failure?

Synchronized Cardioversion*

- Consider sedation
- If regular narrow complex, consider adenosine

Wide QRS?
0.12 second

- IV access and 12-lead ECG if available
- Consider adenosine only if regular and monomorphic
- Consider antiarrhythmic infusion
- Consider expert consultation

- IV access and 12-lead ECG if available
- Vagal maneuvers
- Adenosine (if regular)
- β -Blocker or calcium channel blocker
- Consider expert consultation

Doses/Details

Synchronized Cardioversion**

Initial recommended doses:

- Narrow regular: 50–100 J
- Narrow irregular: 120–200 J biphasic or 200 J monophasic
- Wide regular: 100 J
- Wide irregular: Defibrillation dose (not synchronized)

Adenosine IV Dose:

First dose: 6 mg rapid IV push; follow with NS flush.
Second dose: 12 mg if required

Antiarrhythmic Infusions for Stable Wide-QRS Tachycardia Procainamide IV Dose:

20–50 mg/min until arrhythmia suppressed, hypotension ensues, QRS duration increases $> 50\%$ or maximum dose 17 mg/kg given.
Maintenance infusion: 1–4 mg/min.
Avoid if prolonged QT or CHF.

Amiodarone IV Dose:

First dose: 150 mg over 10 minutes.
Repeat as needed if VT recurs. Follow by maintenance infusion of 1 mg/min for first 6 hours.

Sotalol IV Dose:

100 mg (1.5 mg/kg) over 5 minutes.
Avoid if prolonged QT.

Simulation Scenario Template

Appendix C: Facilitator Cheat Sheet & Debriefing Tips

Objectives:

- Recognition of wide complex tachycardia
- Following the stable tachycardia ACLS algorithm
- Recognition of unstable tachycardia with a pulse
- Differential diagnosis of wide complex tachycardia
- Communicating clearly with team in during an unstable cardiac resuscitation
- Closed loop communication

Key moments to reflect on:

- Identification of WCT
- Discussion or recognition of the differential of WCT, including pre-excitation
- Recognition of when patient turned unstable
- Performing electrical cardioversion

Sample debriefing questions:

- What was everyone's understanding of the rhythm that the patient was in and its management?
- What memory aids could you use to reduce cognitive load?

References

1. [ACLS algorithm 2020](#)
2. [Life in the fast lane](#)
3. [ACC.org](#)

